

Midterm Mastery (Part II)

Start Assignment

- Due Monday by 12pm
- Points 10
- Submitting a text entry box, a website url, or a file upload

Note: this picks up where Part I left off :)

Step III: Crashing and recovering!

Well, systems certainly do crash... Let's lean into that and capture an example that you can creatively construct on your own to demonstrate a sample application that crashes (or fails in some fun way?!). The application can be based on one of our assignments, or one of your own! However, it will be important to **demonstrate** that you have a way to fix the problem too. Sam Newman's overview of approaches like (1) fail fast, (2) circuit breaker, and (3) bulkhead would be great to "show off" in your example, but there would be many more ways you could address the problem in your system. Again, feel free to be creative, and surprise your group with your results! More specifically, be prepared to show your code and demo this result in your group, as follows:

1. the deployment that is problematic, including metrics/evaluation of demonstrating the issue
2. the overview of the approach/code you used to fix it!
3. the improved system running with your fix, including the metrics/evaluation demonstrating the fix

Stay tuned, as we will be preparing a "Midterm Mystery" application for you to explore as well! (more on this soon!)

Step IV: Presentation

Prepare an 8-10 minute presentation for your group, based on Steps I - III of this Midterm Mastery, but also be ready to hop over to Piazza to show your posting on your project (below)! Also be sure to get the contact information for your mock interview group, as you will team up for the *next* assignment, and maybe even the final project! :)

Step V: Project Proposal

We are very excited to hear what you would like to do for your project, which will be a deeper dive into something you are excited about! For now, share your outline on Piazza, and maybe we can get some groups working on common themes. Your posting should contain the following:

- ○ A description of the codebase you want to build, and the ways it needs to scale
- ○ An outline at least 3 experiments you want to do to evaluate the scalability of the system

- If you are looking for group members to join you on this amazing journey (maybe include your time zone too!)

What to submit!

For **Monday, Mar 2 9am Pacific Time**

- Step III, also submit a separate report for your experiment. Include screen shots, and charts/graphs for the evaluation of your amazing crashing and recovering experiment! NOTE: you can be very creative here! Surprise your team :)
- Step IV, please upload a .pdf of the slides you used in your short presentation.
- Step V, this will be a post on Piazza. Feel free to reach out to other people who are posting on topics you are interested in. You are encouraged to team up with your mock interview group, but other groupings are also possible. Just make sure you are working on a topic you are excited about!

Mock Interview Scoring! (5)

Criteria	Ratings					Pts
Code Quality	2 pts Full Marks Excellent! Very clear and concise, outstanding documentation, readability, and maintainability. Beautiful code!	1.5 pts Good quality Clarity could be improved, in particular documentation.	1 pts Could use improvement Code readability and maintainability could be improved, this will come with practice!	0.5 pts Major improvement expected Code quality is poor, not something that could easily be shared with a team in a code walk.	0 pts No code submitted by due date.	2 pts
Code Completion	2 pts Full Marks Excellent! Code is complete and runs without error, thought has been put into testing and error checking.	1.5 pts Small bugs Minor runtime errors, cleared up in interview time, but otherwise code runs correctly.	1 pts Needs improvement Code does not run and suffers from major errors, and requires further testing.	0.5 pts Needs major improvement Some code was submitted, but has significant errors and has not been subjected to adequate testing.	0 pts No Code was submitted by due date.	2 pts
Code understanding	2 pts Full Marks Excellent job in answering questions about the code and analysis posed by both the TA and other group members!	1.5 pts Good answers Answers to questions were clear, but there is room to improve and go deeper into technical explanations.	1 pts Communication improvements expected Ability to show a deeper understanding is expected.	0.5 pts Major Communication Issue Answering questions will require more practice, both in terms of listening carefully to the question that is asked, and thinking deeply about the tradeoffs in the answers.	0 pts No Code was submitted by due date.	2 pts

Criteria	Ratings					Pts
Listening and Engaging with team	2 pts Full Marks Excellent job of contributing to the group discussion! Listening closely to others, asking insightful questions, helping consider alternatives!	1.5 pts Strong interaction Some elements of group interaction can be improved, in particular making questions/comments specific and insightful.	1 pts Some improvements expected Ensure listening is uninterrupted and the discussion is progressing. Do not repeat questions/comments that have already been shared.	0.5 pts Major Engagement Issue Did not appear to be listening, and was unable to contribute beyond superficial comments.	0 pts No questions asked No interaction with team members	2 pts
Understanding of Concepts and Tradeoffs	2 pts Full Marks Excellent job of mapping concrete code examples to high level concepts! This was demonstrated throughout the interview, and made abundantly clear in the posting on the reading on Piazza!	1.5 pts Good sense of tradeoffs Is able to see the connection to higher level concepts, and is working on communicating the relationships between code examples and concepts. Posting on Piazza was strong.	1 pts Room for improvement With practice, the ability to reason about concepts and tradeoffs and communicate clear points will be improved. Developing this new skill, posting on Piazza showed promise.	0.5 pts Needs Major improvement Posting on Piazza needs more depth. Currently not connecting examples to concepts and tradeoffs, but aware of the nature of this relationship.	0 pts No Marks No posting on the reading on Piazza.	2 pts
Total Points: 10						